

Notes: Zhang, Zhu, and Linnainmaa (RFS, 2025)

Man versus Machine Learning Revisited

Contents

1	Big picture	3
2	Data and measurement (key objects)	4
2.1	Paper being revisited	4
2.2	Data sources	4
2.3	Sample	4
2.4	Features used to forecast earnings	5
2.5	The look-ahead feature	5
2.6	Replication implementation choices	6
2.7	Key constructed variables	6
2.8	Outcomes	6
3	Models and empirical specifications (all equations + interpretation)	7
3.1	Random-forest earnings forecast	7
3.2	Correct versus look-ahead timing	7
3.3	Conditional analyst forecast bias	8
3.4	Portfolio strategy	8
3.5	Factor-alpha regressions	9
3.6	Forecast-accuracy comparisons	9
3.7	Direct diagnostic for the look-ahead bias	9
3.8	Decomposing conditional bias	10
3.9	Alternative machine-learning models	10
4	Identification and instruments (what makes the estimates credible)	11
4.1	What fails in the original design	11
4.2	What identifies the critique	11

4.3	Why the replication is credible	11
4.4	What the paper does not claim	12
4.5	Relation to the literature	12
5	Tables and figures	12
5.1	Table 1: five-factor alphas for portfolios sorted on conditional bias	12
5.2	Table 2: replication details	12
5.3	Table 3: term structure of earnings forecasts	13
5.4	Table 4: average returns of bias-sorted portfolios	13
5.5	Figure 1: Sharpe ratios of anomaly portfolios	14
5.6	Figure 2: cumulative returns of bias-sorted strategies	14
5.7	Table 5: factor-model alphas under CAPM, FF3, and FF5	15
5.8	Table 6: alternative modern factor models	15
5.9	Table 7: comparison with So (2013)	16
5.10	Look-ahead diagnostic regressions	16
5.11	Other BHL results: equity issuance and anomaly returns	16
5.12	Alternative machine-learning forecasts	17
6	Short summary	17
7	Conclusion	18
7.1	Core conclusion	18
7.2	Economic intuition	18
7.3	Takeaway	18

1 Big picture

- **Question.** Does the machine-learning benchmark in Binsbergen, Han, and Lopez-Lira (2023), abbreviated BHL, really uncover economically large and tradable biases in sell-side analysts' earnings forecasts?
- **Setting.** The paper revisits BHL's random-forest forecast of firms' future earnings and the trading strategy that sorts stocks on the gap between analysts' forecasts and the random-forest benchmark.
- **What BHL claimed.** BHL train random forests to forecast earnings over five horizons—one quarter, two quarters, three quarters, one year, and two years ahead. They define analysts' conditional bias as analyst forecast minus the machine-learning benchmark. A value-weighted strategy that trades against this bias earns a monthly Fama–French five-factor alpha of about 1.54% in absolute value, with $t = 5.84$.
- **Why this matters.** If correct, the BHL result would be unusually strong:
 - it would imply that machine learning dramatically improves the measurement of analysts' forecast biases;
 - it would set a very high benchmark for new return-predictability papers;
 - it would encourage later researchers to use the BHL forecast output as a standard control or benchmark.
- **Core problem found by Zhang, Zhu, and Linnainmaa.** One feature in the BHL random forest is “realized earnings from the last period.” For some forecast horizons, this is not actually known at the forecasting date. For example, when forecasting two-year-ahead earnings, the “last-period” realized earnings are one-year-ahead earnings, which are future information at time t .
- **Main contribution.** The paper reverse-engineers BHL's specification, replicates the original result when the look-ahead variable is included, then removes only the look-ahead timing error and studies what remains.
- **Main empirical result.**
 - With the look-ahead bias, the authors replicate BHL closely: the five-factor alpha is -1.71% per month, compared with BHL's -1.54% .
 - Without the look-ahead bias, the five-factor alpha falls to -0.41% with $t = -1.54$, so it is not statistically significant.
 - Under five alternative modern factor models, the corrected strategy earns alphas close to zero, with t -statistics between about -0.27 and 0.87 .
 - The raw high-minus-low return spread falls from about -1.59% per month with the look-ahead bias to only -0.25% without it.
- **Forecasting result.** Removing the look-ahead bias also reduces forecast accuracy, especially at the affected horizons $Q2$, $Q3$, and $A2$. A simpler linear model using analyst forecasts and classical predictors predicts earnings about as accurately as the corrected random forest.
- **Broader result.** The look-ahead bias does not overturn every BHL result, but it changes the interpretation:

- equity issuance and anomaly-short-leg results are not very sensitive to the bias, but they appear to be driven by the expected-earnings component rather than by analyst optimism;
- the result that anomalies are stronger among high-bias stocks remains, and here the random forest adds some value;
- alternative machine-learning models do not resurrect the large alpha.
- **Economic takeaway.** Machine learning can be useful for earnings forecasts, but the BHL alpha is not evidence that machine learning automatically uncovers large tradable analyst-bias mispricing. Timing discipline is central: a small feature-timing mistake can create a very large apparent anomaly.

2 Data and measurement (key objects)

2.1 Paper being revisited

- The target paper is Binsbergen, Han, and Lopez-Lira (2023), *Man versus Machine Learning: The Term Structure of Earnings Expectations and Conditional Biases*.
- BHL’s object is an expected-earnings benchmark produced by random forests. Analysts’ forecast bias is measured by comparing analysts’ EPS forecasts with that benchmark.
- Zhang, Zhu, and Linnainmaa ask whether BHL’s forecast benchmark and trading strategy survive a stricter treatment of information timing.

2.2 Data sources

- The replication uses the same four main sources as BHL:
 - monthly stock-level data from **CRSP**;
 - analysts’ consensus earnings forecasts and actual earnings from **IBES Unadjusted Summary** and **IBES Unadjusted Detail**;
 - accounting and firm-characteristic data from the **WRDS Financial Ratios Suite**;
 - real-time macroeconomic indicators from the **Federal Reserve Bank of Philadelphia**.
- BHL also provide their original conditional-bias estimates, which Zhang, Zhu, and Linnainmaa use for direct comparison and for the look-ahead diagnostic tests.

2.3 Sample

- The main sample covers U.S. common stocks:
 - CRSP share codes 10 and 11;
 - NYSE, AMEX, and Nasdaq exchange codes 1, 2, and 3.
- The main forecasting sample runs from **January 1984 through December 2019**.
- The return-predictability tests generally run from **January 1986 through December 2019**, matching BHL’s trading-sample timing.

- The paper studies five forecast horizons:

$$h \in \{Q1, Q2, Q3, A1, A2\},$$

where $Q1, Q2, Q3$ are one-, two-, and three-quarter-ahead EPS forecasts, and $A1, A2$ are one- and two-year-ahead EPS forecasts.

2.4 Features used to forecast earnings

- BHL use roughly 75 features, grouped into three categories.
- **Firm characteristics:**
 - realized earnings from the “last period”;
 - monthly stock prices and returns from CRSP;
 - 67 financial ratios from WRDS Financial Ratios Suite;
 - missing values filled with Fama–French 49-industry medians.
- **Macroeconomic variables:**
 - consumption growth;
 - real GDP growth;
 - industrial production growth;
 - unemployment rate.
- **Analyst forecasts:**
 - analysts’ quarterly and annual consensus EPS forecasts.
- **Key measurement issue.** The first firm-characteristic feature—realized earnings from the “last period”—is the look-ahead variable when it is defined relative to the target earnings date instead of the forecasting date.

2.5 The look-ahead feature

- BHL estimate separate random forests for the five forecast horizons. The problem is that the “last-period realized earnings” feature is horizon-specific.
- For the $Q1$ model, last-period realized earnings are genuinely prior-quarter earnings and are known at time t .
- For the $Q2$ model, last-period realized earnings are $Q1$ earnings, which are not yet known at time t .
- For the $Q3$ model, last-period realized earnings are $Q2$ earnings, again unknown at time t .
- For the $A2$ model, last-period realized earnings are $A1$ earnings, also unknown at time t .
- Thus three of the five models— $Q2$, $Q3$, and $A2$ —contain future realized earnings as a feature.
- Zhang, Zhu, and Linnainmaa correct this by using realized earnings from the most recent period available as of the forecasting date.

2.6 Replication implementation choices

- BHL’s paper and posted code leave some implementation details unspecified or inconsistent, so the authors search over plausible choices and choose the specification that best matches BHL or is simplest when results are similar.
- Main implementation choices:
 - **standardization:** try cross-sectional z -scores, ranks, and none; the main specification uses no standardization because it best matches BHL;
 - **winsorization:** winsorize both features and targets, because this better matches the posted code;
 - **per-share variables:** exclude additional per-share variables from the main specification because including them gives similar results and their exact BHL list is unclear;
 - **sample fraction:** use 5%, matching the posted code rather than the 1% stated in BHL’s text;
 - **feature fraction:** use all features at each split;
 - **number of trees:** 2,000;
 - **maximum tree depth:** 7;
 - **minimum node size:** 5;
 - **training window:** 12 months for quarterly and one-year-ahead forecasts, 24 months for two-year-ahead forecasts;
 - **feature timing:** analyst forecasts are available in month t ; all other features should be available by the end of month $t - 1$.

2.7 Key constructed variables

- $^h_{i,t}$: realized EPS for firm i at horizon h relative to month t .
- $\text{AF}^h_{i,t}$: analyst consensus EPS forecast for firm i at horizon h in month t .
- $\text{RF}^h_{i,t}$: random-forest expected-earnings benchmark for firm i , month t , and horizon h .
- $\text{BE}^h_{i,t}$: conditional bias estimate, defined as analyst forecast minus random-forest benchmark, scaled by lagged price.
- $\overline{\text{BE}}_{i,t}$: average conditional bias across the five horizons, computed when at least two horizon-specific bias estimates are nonmissing.
- R_t^{5-1} : the value-weighted return on the high-bias-minus-low-bias portfolio. Since high bias means more optimistic analysts relative to the benchmark, the sign-reversed low-minus-high strategy is the natural “trade against analyst bias” strategy.

2.8 Outcomes

- **Forecast accuracy:** mean forecast errors and mean-squared forecast errors, especially $(\text{RF} -)^2$ versus $(\text{AF} -)^2$.

- **Return predictability:** average returns and factor alphas for portfolios sorted on $\overline{\text{BE}}_{i,t}$.
- **Robustness to factor models:** CAPM, Fama–French three-factor, Fama–French five-factor, Fama–French–Carhart six-factor, Hou–Xue–Zhang q -factor, Hou et al. $q5$, Stambaugh–Yuan mispricing, and Daniel–Hirshleifer–Sun behavioral factors.
- **Other BHL outcomes:** equity issuance, anomaly short-leg classification, and anomaly performance conditional on estimated bias.

3 Models and empirical specifications (all equations + interpretation)

3.1 Random-forest earnings forecast

For each month t and horizon h , the random forest predicts future earnings:

$$\mathbb{E}_t \left[\overset{h}{\underset{i,t}{\text{EPS}}} \right] = f_t^h(X_{i,t} \mid \gamma),$$

where $\overset{h}{\underset{i,t}{\text{EPS}}}$ is realized EPS at horizon h , $X_{i,t}$ is the feature vector, and γ collects random-forest hyperparameters.

Interpretation.

- The random forest is meant to be a data-driven benchmark for what earnings should be expected to be, given observable information at time t .
- If the features are truly known at t , the forecast can be interpreted as an implementable benchmark.
- If any feature uses information realized after t , the benchmark is contaminated by look-ahead bias.

3.2 Correct versus look-ahead timing

Let $\text{LastEarn}_{i,t}^h$ denote the realized-earnings feature used in the horizon- h random forest.

The look-ahead construction is effectively

$$\text{LastEarn}_{i,t}^{h,LA} = \overset{h-1}{\underset{i,t}{\text{EPS}}},$$

while the correct construction should be

$$\text{LastEarn}_{i,t}^{h,C} = \text{LatestKnownEarnings}_{i,t}.$$

For example, when $h = A2$,

$$\text{LastEarn}_{i,t}^{A2,LA} = \overset{A1}{\underset{i,t}{\text{EPS}}},$$

which is the firm's one-year-ahead realized EPS, not information available to an investor at month t .

Interpretation.

- The random forest is not merely learning from past earnings; for $Q2$, $Q3$, and $A2$, it is partly learning from future earnings.
- This makes the machine-learning benchmark mechanically closer to the future realized earnings.
- The look-ahead bias therefore improves measured forecast accuracy and can create spurious return predictability.

3.3 Conditional analyst forecast bias

For each horizon, BHL's conditional bias estimate is

$$\text{BE}_{i,t}^h = \frac{\text{AF}_{i,t}^h - \text{RF}_{i,t}^h}{\text{Price}_{i,t-1}}.$$

The average conditional bias is

$$\overline{\text{BE}}_{i,t} = \frac{1}{|\mathcal{H}_{i,t}|} \sum_{h \in \mathcal{H}_{i,t}} \text{BE}_{i,t}^h,$$

where $\mathcal{H}_{i,t} \subseteq \{Q1, Q2, Q3, A1, A2\}$ is the set of horizons with nonmissing bias estimates. BHL require at least two nonmissing horizons.

Interpretation.

- A high BE means analysts' EPS forecast is high relative to the machine-learning benchmark.
- A low BE means analysts' EPS forecast is low relative to the machine-learning benchmark.
- If analysts' optimism causes overpricing, high-bias stocks should underperform and low-bias stocks should outperform.

3.4 Portfolio strategy

Each month, stocks are sorted into quintiles by $\overline{\text{BE}}_{i,t}$. The reported long-short return is the high-bias-minus-low-bias portfolio:

$$R_t^{5-1} = R_t^{Q5} - R_t^{Q1}.$$

The trade-against-bias strategy is the sign-reversed portfolio:

$$R_t^{1-5} = R_t^{Q1} - R_t^{Q5} = -R_t^{5-1}.$$

Interpretation.

- BHL report large negative returns and alphas for R_t^{5-1} .

- Economically, this means the low-bias portfolio outperforms the high-bias portfolio.
- The sign is less important than the magnitude and statistical significance of the spread.

3.5 Factor-alpha regressions

For the Fama–French five-factor model, the main alpha regression is

$$R_t^{5-1} = \alpha + b_M \text{MKT}_t + b_S \text{SMB}_t + b_H \text{HML}_t + b_R \text{RMW}_t + b_C \text{CMA}_t + \varepsilon_t.$$

The paper also estimates analogous models under other factor structures.

Interpretation.

- α measures abnormal return after controlling for standard risk and characteristic-based factor exposures.
- BHL’s main evidence is that α is large and highly significant.
- Zhang, Zhu, and Linnainmaa show that the corrected α is not significant under modern factor models.

3.6 Forecast-accuracy comparisons

Forecast accuracy is summarized by horizon-specific mean-squared errors:

$$\text{MSE}_h^{RF} = \mathbb{E} \left[(\text{RF}_{i,t}^h - \frac{h}{i,t})^2 \right], \quad \text{MSE}_h^{AF} = \mathbb{E} \left[(\text{AF}_{i,t}^h - \frac{h}{i,t})^2 \right].$$

The paper also compares the corrected random forest to a linear model:

$$\text{MSE}_h^{RF,C} - \text{MSE}_h^{OLS}.$$

Interpretation.

- The look-ahead-biased random forest appears more accurate partly because it sees future earnings.
- After correcting the timing, the random forest still improves on analysts at the five horizons, but it no longer significantly improves on a simpler OLS model that includes analyst forecasts and standard accounting predictors.
- The most important feature for accuracy is not the nonlinear algorithm itself; it is the inclusion of analysts’ forecasts as an input.

3.7 Direct diagnostic for the look-ahead bias

The paper directly tests whether BHL’s longer-horizon forecasts contain future realized earnings. The diagnostic regression is

$$\frac{h}{i,t} = \alpha + \beta_1 \text{FE}_{i,t}^h + \beta_2 \text{FE}_{i,t}^{h+1} + \gamma_1 \text{AF}_{i,t}^h + \gamma_2 \text{AF}_{i,t}^{h+1} + \varepsilon_{i,t}.$$

If the horizon- $h + 1$ forecast embeds horizon- h realized earnings, then

$$FE_{i,t}^{h+1} = \hat{f}_t^{h+1}(^h_{i,t}, \dots),$$

so the regression should produce a positive β_2 , even after controlling for the horizon- h forecast and analyst forecasts.

Interpretation.

- This is a direct test of the timing error.
- If there is no look-ahead bias, the $(h + 1)$ -horizon machine forecast should not have special power to predict h -horizon realized earnings once the h -horizon forecast and analyst forecasts are controlled for.
- In BHL's provided forecasts and in the authors' look-ahead replication, $\hat{\beta}_2$ is strongly positive, with t -statistics as high as about 30.2.
- When the look-ahead bias is removed, this diagnostic pattern disappears.

3.8 Decomposing conditional bias

The BHL bias variable can be written as

$$\text{Conditional Bias}_{i,t} = \text{Analyst Forecast}_{i,t} - \text{Expected Earnings Benchmark}_{i,t}.$$

Interpretation.

- A high conditional bias can mean that analysts are especially optimistic.
- But it can also mean that the model's expected earnings benchmark is especially low.
- This distinction matters for interpreting the equity issuance and anomaly-short-leg results: if those results are driven by low predicted earnings, they are not direct evidence about analyst optimism.

3.9 Alternative machine-learning models

The paper also evaluates alternative forecasting approaches:

$$\mathcal{F}_{i,t}^h \in \{\text{OLS, partial least squares, LASSO, elastic net, random forest, LightGBM, composite}\}.$$

Interpretation.

- The composite model averages forecasts across several methods and produces the best overall forecast accuracy.
- But better earnings forecasts do not automatically translate into profitable return signals.
- None of the alternative models restores the large BHL alpha.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in the original design

- The BHL empirical design relies on the assumption that all random-forest features are observable at the forecasting date.
- If future realized earnings enter the feature set, the model is not an implementable forecast model. It becomes partly a hindsight-based prediction device.
- The key failure is not the use of machine learning itself. The failure is an information-set error in the data construction.

4.2 What identifies the critique

- The critique is identified by comparing three objects:
 - BHL’s original reported results;
 - the authors’ replication with the look-ahead variable intentionally included;
 - the same replication pipeline with only the timing of the realized-earnings feature corrected.
- Because the replication with the bias closely matches BHL, the authors show that they have found the relevant implementation.
- Because removing the look-ahead feature collapses the alpha, the authors show that the large alpha is driven by the timing error rather than by robust machine-learning discovery.

4.3 Why the replication is credible

- **Close match to BHL.** With the look-ahead variable included, the replication delivers a five-factor alpha of -1.71% , close to BHL’s -1.54% , and factor loadings are also close.
- **Single correction.** The main correction changes the realized-earnings feature to information available at the forecast date while keeping the rest of the pipeline fixed.
- **Affected horizons line up.** The degradation occurs at $Q2$, $Q3$, and $A2$, exactly the horizons where the feature contains future earnings.
- **Direct diagnostic.** BHL’s longer-horizon machine forecast predicts shorter-horizon realized earnings in the way expected if the longer-horizon forecast embeds the shorter-horizon actual.
- **Alternative factor models.** The corrected strategy fails under several modern factor models, not merely under one specification.
- **Classical-model benchmark.** A linear model using standard variables performs as well as the corrected random forest in forecast accuracy, weakening the claim that nonlinear machine learning is the source of the original result.
- **Alternative ML benchmark.** Other machine-learning methods and a composite forecast do not resurrect the return-predictability result.

4.4 What the paper does not claim

- The paper does not claim that analysts' forecasts are always unbiased.
- It does not claim that machine learning is useless for earnings forecasts.
- It does not claim that the term structure of analyst expectations is irrelevant for valuations.
- Its narrower claim is that the unusually large BHL trading alpha and some interpretations of BHL's evidence depend on a look-ahead bias.

4.5 Relation to the literature

- **Analyst forecast bias.** The paper connects to La Porta (1996), So (2013), Hughes, Liu, and Su (2008), and Kothari, So, and Verdi (2016), which study whether analyst forecasts embed predictable errors and whether those errors are priced.
- **Machine learning in finance.** The paper contributes a cautionary lesson: out-of-sample prediction in finance requires strict information-set discipline, especially when horizons differ.
- **Asset-pricing replication.** The paper is also an example of careful replication and forensic specification checking: it shows how a strong anomaly can arise from a subtle data-timing error.

5 Tables and figures

5.1 Table 1: five-factor alphas for portfolios sorted on conditional bias

Read:

- BHL's original high-minus-low bias portfolio has a five-factor alpha of -1.54% per month with $t = -5.84$.
- The authors' replication with the look-ahead bias gives -1.71% with $t = -6.07$, so they reproduce the original result very closely.
- Removing the look-ahead bias reduces the alpha to -0.41% with $t = -1.54$.
- The key message is that the strong alpha is not robust to correcting the timing of realized earnings.

5.2 Table 2: replication details

Read:

- The table documents the ambiguous choices in BHL and the choices used in the replication.
- The most important row is the timing of last-period realized earnings:

with look-ahead: $t + h - 1$, without look-ahead: $t - 1$.

Table 1
Monthly Fama-French five-factor model alphas and factor loadings of portfolios sorted on analysts' conditional biases

	Replication					
	BHL Original		With Look-Ahead Bias		Without Look-Ahead Bias	
	Coef.	<i>t</i> -value	Coef.	<i>t</i> -value	Coef.	<i>t</i> -value
Intercept	-1.54	-5.84	-1.71	-6.07	-0.41	-1.54
MKT	0.38	5.28	0.43	5.71	0.44	5.94
SMB	0.61	5.17	0.66	5.14	0.75	5.69
HML	0.95	7.12	0.90	6.27	0.84	5.90
RMW	-0.68	-4.10	-0.75	-4.01	-0.72	-3.83
CMA	-0.53	-1.93	-0.30	-1.09	-0.18	-0.67

This table reports Fama-French five-factor model alphas and factor loadings for three long-short strategies. Each strategy is long stocks with the highest conditional earnings forecast biases and short stocks with the lowest biases. We measure the conditional biases in three steps, following [Binsbergen, Han, and Lopez-Lira \(2023\)](#). First, we measure the expected earnings benchmarks over the 1-quarter, 2-quarter, 3-quarter, 1-year, and 2-year horizons using random forest regressions. Second, we compute the differences between analysts' consensus earnings forecasts and these benchmarks at each horizon. Finally, we average these differences at the five horizons as the conditional bias. Columns 1 and 2 report the estimates from Table 6 in [Binsbergen, Han, and Lopez-Lira \(2023\)](#). Columns 3 and 4 show our replication of the result. Columns 5 and 6 remove the variable with the look-ahead bias from the random forest regression used to estimate the conditional biases. The sample period is January 1986 through December 2019.

Table 2
Replication details

Panel A: Choices experimented		
	BHL	Replication
Timing of last-period realized earnings	Unspecified	With look-ahead: $t+h-1$
Standardization	Unspecified	Without look-ahead: $t-1$
		Z-scores
		Ranks
Winsorization	Features (text)	None
	Features and targets (code)	Features
Per-share data	Unspecified	Features and targets
		Include
		Exclude
Sample fraction	1% (text)	1%
	5% (code)	5%
Feature fraction	Top five features (text) plus 50% at random (code)	Top five features plus 50% at random
		All features
Panel B: Other details		
Feature timing	Analysts' forecast at time t and all other features at $t-1$	
Number of trees		2,000
Maximum depth		7
Minimum node size		5
Training window	24 months for year-2 forecasts and 12 months for other forecasts	

This table reports the specifications of the replication. Panel A reports which details are unspecified in BHL and the set of choices tried. The bolded option corresponds to the main specification, chosen on the basis of providing a better match with the results in BHL and/or simplicity. Panel B reports the other implementation details provided in BHL.

- Other choices include no standardization, winsorization of features and targets, a 5% sample fraction, all features at each split, 2,000 trees, maximum depth 7, and minimum node size 5.
- The table shows that the authors are not changing the research design broadly; they are isolating implementation details, especially timing.

5.3 Table 3: term structure of earnings forecasts

Read:

- The authors closely replicate BHL's machine forecasts and analyst forecast errors when the look-ahead bias is included.
- Analysts' forecasts are upward biased across horizons: $AF -$ is positive and statistically significant.
- Removing the look-ahead bias worsens random-forest accuracy at $Q2$, $Q3$, and $A2$, the affected horizons.
- The largest effect is for $A2$. With the look-ahead bias, the random forest reduces mean-squared error by 0.213 relative to analysts. Without the bias, the improvement shrinks to 0.051. Roughly three-quarters of the apparent advantage comes from the look-ahead bias.
- The degradation is statistically large: the $A2$ deterioration has $t = -10.50$.
- The corrected random forest does not significantly outperform the OLS model at any horizon.

5.4 Table 4: average returns of bias-sorted portfolios

Read:

- BHL's high-minus-low return spread is -1.46% per month with $t = -5.11$.
- The authors' look-ahead replication gives -1.59% with $t = -5.08$.
- Without the look-ahead bias, the spread falls to -0.25% with $t = -0.80$.
- Thus the raw return spread collapses by about four-fifths after the correction.

Table 3
The term structure of earnings forecasts via machine learning

	(1) RF	(2) AF	(3) AE	(4) RF-AE	(5) r(RF-AE)	(6) AF-AE	(7) r(AF-AE)	(8) r(RF-AE) ²	(9) r(AF-AE) ²	(10) N
Panel A: BHL original (Table 2)										
Q1	0.290	0.319	0.291	-0.090	-0.17	0.028	6.59	0.076	0.081	1,022,661
Q2	0.323	0.376	0.323	-0.091	-0.13	0.053	10.31	0.094	0.102	1,110,689
Q3	0.343	0.413	0.341	0.002	0.31	0.072	11.25	0.121	0.132	1,010,908
A1	1.194	1.320	1.167	0.027	1.64	0.154	6.24	0.070	0.086	1,260,090
A2	1.387	1.771	1.387	-0.004	-0.07	0.384	8.33	1.897	2.009	1,097,098
Panel B: Replication with look-ahead bias										
Q1	0.288	0.317	0.289	-0.001	-0.37	0.028	6.50	0.071	0.078	1,024,654
Q2	0.319	0.374	0.323	-0.004	-0.83	0.051	10.17	0.087	0.096	1,118,537
Q3	0.338	0.411	0.341	-0.002	-0.40	0.070	11.38	0.111	0.125	1,026,893
A1	1.184	1.306	1.157	0.027	1.89	0.149	6.14	0.062	0.075	1,265,472
A2	1.349	1.727	1.359	-0.011	-0.20	0.368	8.26	1.643	1.856	1,136,233

Panel C: Replication without look-ahead bias and accuracy comparison

	Replication without look-ahead bias				Accuracy improvement in RF-AE ²			
	RF	RF-AE	r(RF-AE)	(RF-AE) ²	Against look-ahead forecast		Against linear model forecast (Hagihara, Lin, and Su 2009)	
	RF	RF-AE	r(RF-AE)	(RF-AE) ²	Difference	r-value	Difference	r-value
Q1	0.288	-0.001	-0.37	0.071	-0.003	-6.48	0.002	1.30
Q2	0.318	-0.004	-0.95	0.090	-0.009	-6.84	0.003	1.10
Q3	0.337	-0.003	-0.51	0.139	-0.009	-6.84	0.004	0.91
A1	1.184	0.027	1.89	0.602	-0.005	-0.63	-0.003	-0.05
A2	1.356	-0.004	-0.07	1.805	-0.161	-10.50	-0.003	-0.05

Panel A and B report time-series averages of machine learning EPS forecasts (RF), analyst EPS forecasts (AF), and realized earnings (AE), and the time-series averages of the cross-sectional mean-squared forecasting errors by random forest, $(RF - AE)^2$, and by analysts, $(AF - AE)^2$. N denotes the number of observations. We also report the Newey and West (1987) adjusted t -values of the difference between earnings forecasts and realized earnings. The standard errors for quarterly forecasts are adjusted with 3 lags and those for annual forecasts are adjusted with 12 lags. Panel A reports estimates from Table 2 in BHL. Panel B shows the replication results with the look-ahead bias. The first four columns in panel C report the replication results without the look-ahead bias. The look-ahead bias affects the bolded estimates. The next two columns estimate the accuracy improvement from the look-ahead bias. Accuracy improvement is the time-series difference in the cross-sectional mean-squared forecasting errors. The last two columns estimate the accuracy improvement over an OLS model that uses Hagihara, Lin, and Su's (2008) variables and analysts' forecasts for the corresponding horizon. The t -values use Newey-West adjusted standard errors. The sample starts in January 1986 and ends in December 2019.

Table 4
Portfolios sorted on conditional bias

	Conditional bias (average BE) quintile					
	1	2	3	4	5	5-1
BHL original (Table 5)	1.32 (6.53)	0.98 (4.53)	0.79 (3.18)	0.47 (1.62)	-0.14 (-0.35)	-1.46 (-5.11)
Replication with look-ahead bias	1.38 (6.74)	0.92 (4.23)	0.61 (2.44)	0.24 (0.78)	-0.21 (-0.51)	-1.59 (-5.08)
Replication without look-ahead bias	0.99 (4.83)	0.90 (4.10)	0.91 (3.62)	0.85 (2.78)	0.75 (1.81)	-0.25 (-0.80)

This table reports time-series average returns (in percent) for value-weighted monthly rebalanced portfolios formed on analysts' average conditional earnings forecast bias quintiles. The first and second rows are from Table 5 in BHL. The remaining four rows report our replication results with and without the look-ahead bias. The t -values (in parentheses) use White's (1980) heteroskedasticity robust standard errors, as in BHL. The sample starts in January 1986 and ends in December 2019.

5.5 Figure 1: Sharpe ratios of anomaly portfolios

Read:

- The figure compares BHL's strategy with 151 anomalies from Chen and Zimmermann's database under common portfolio rules.
- With the look-ahead bias, the BHL strategy has an annualized Sharpe ratio of about 0.95, higher than all comparable anomalies in the database.
- Without the look-ahead bias, the strategy's Sharpe ratio falls to about 0.15.
- The figure visually shows why the original result was so influential: it looked like an extreme outlier in the anomaly distribution. The corrected result is not an outlier.

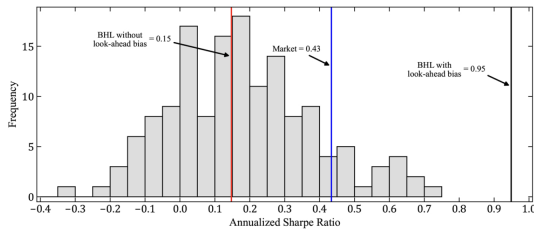


Figure 1
Sharpe ratios of 151 anomaly portfolios

This figure plots the distribution of annualized Sharpe ratios of 151 anomalies from Chen and Zimmermann's (2022) database. We impose the following restrictions both on the BHL strategy and all anomalies. The sample consists of common shares traded in NYSE, AMEX, and Nasdaq that have a positive book value of equity in the prior year and a share price above one dollar in the prior month. We also exclude microcap stocks, defined as those with a market value of equity below the 20th NYSE percentile in the prior month. We require the anomaly variables to be continuous (that is, excluding discrete and categorical variables) and available from 1986 to 2019 for at least 500 stocks in the average month. We augment the Chen-Zimmermann database with the price, size, and short-term reversal signals that it omits. All anomalies are value-weighted long-short portfolios based on monthly rebalanced quintiles. The black, blue, and red vertical lines indicate the Sharpe ratios of the BHL strategy (with the look-ahead bias), the market portfolio, and our replication of BHL (without the look-ahead bias), respectively. The Sharpe ratio of the market portfolio is computed from July 1926 to December 2019.

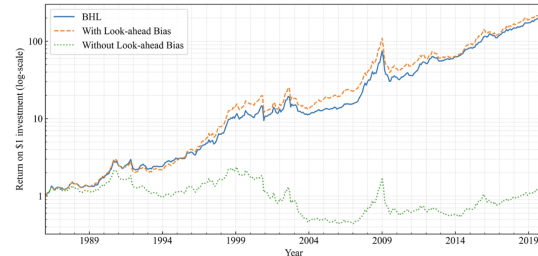


Figure 2
Cumulative returns of value-weighted portfolios formed on conditional bias

This figure shows cumulative returns for monthly rebalanced value-weighted strategies that buy (sell) stocks in the lowest (highest) "Average BE" quintile. We implement this strategy using (a) the estimates provided by BHL (blue solid line), (b) our replication with a look-ahead bias (orange dashed line), and (c) our replication without a look-ahead bias (green dotted line). The sample period begins in January 1986 and ends in December 2019.

5.6 Figure 2: cumulative returns of bias-sorted strategies

Read:

- The cumulative return path using BHL's provided bias estimates and the authors' look-ahead replication move almost together.
- Their monthly return correlation is about 0.95.

- The corrected strategy, without the look-ahead bias, is consistently much weaker and does not generate the same cumulative performance.
- This figure reinforces the replication logic: the authors can reproduce BHL when the future-earnings feature is included, but not when it is removed.

5.7 Table 5: factor-model alphas under CAPM, FF3, and FF5

Read:

- With the look-ahead bias, both BHL's original strategy and the replication earn large and significant alphas under CAPM, Fama–French three-factor, and Fama–French five-factor models.
- Without the bias, the CAPM and FF3 alphas remain negative and statistically significant, but the FF5 alpha becomes insignificant.
- The attenuation under FF5 matters because the remaining profits are related to profitability and investment factor exposures.
- Interpretation: after correcting the timing, the strategy is not a distinct analyst-bias alpha once modern factor exposures are controlled for.

Table 5
Time-series tests with common asset pricing models

	CAPM		FF3		FF5	
	Coef.	<i>t</i> -value	Coef.	<i>t</i> -value	Coef.	<i>t</i> -value
<i>Panel A: BHL original (Table 6)</i>						
Intercept	−1.85	−7.18	−1.96	−8.64	−1.54	−5.84
MKT	0.56	7.53	0.53	7.86	0.38	5.28
SMB			0.80	7.06	0.61	5.17
HML			0.58	5.25	0.95	7.12
RMW					−0.68	−4.10
CMA					−0.53	−1.93
<i>Panel B: Replication with look-ahead bias</i>						
Intercept	−2.00	−7.25	−2.10	−8.63	−1.71	−6.07
MKT	0.61	7.49	0.56	7.56	0.43	5.71
SMB			0.88	7.22	0.66	5.14
HML			0.62	5.35	0.90	6.27
RMW					−0.75	−4.01
CMA					−0.30	−1.09
<i>Panel C: Replication without look-ahead bias</i>						
Intercept	−0.67	−2.45	−0.76	−3.26	−0.41	−1.54
MKT	0.61	7.75	0.55	7.83	0.44	5.94
SMB			0.96	7.82	0.75	5.69
HML			0.61	5.48	0.84	5.90
RMW					−0.72	−3.83
CMA					−0.18	−0.67

This table reports the alphas and factor loadings for the value-weighted long-short portfolio under the CAPM, Fama and French (1993) three-factor (FF3), and Fama and French (2015) five-factor models (FF5). The results in panel A are from BHL's Table 6. Panels B and C present the results using our replicated average conditional bias (BE) with and without look-ahead bias. The *t*-values use White's (1980) heteroskedasticity robust standard errors, as in BHL. The sample starts in January 1986 and ends in December 2019.

Table 6
Alternative models without look-ahead bias

Model	FFC6	HXZ- <i>q</i> 4	HMXZ- <i>q</i> 5	SY	DHS
Intercept	−0.02 (−0.11)	−0.08 (−0.27)	−0.01 (−0.02)	0.07 (0.25)	0.31 (0.87)

This table reports the alphas of the long-short strategy without look-ahead bias under the Fama and French (2015) and Carhart (1997) six-factor model (FFC6), Hou, Xue, and Zhang (2015) *q*-factor model (HXZ), Hou et al. (2021) *q*5-factor model (HMXZ), Stambaugh and Yuan (2016) mispricing factor model (SY), and Daniel, Hirshleifer, and Sun (2020) short- and long-horizon behavioral model (DHS). The *t*-values (in parentheses) use White's (1980) heteroskedasticity robust standard errors as in BHL. The sample starts in January 1986 and ends in December 2019.

5.8 Table 6: alternative modern factor models

Read:

- The corrected strategy's alphas are close to zero under several modern models:
 - Fama–French–Carhart six-factor: -0.02 , $t = -0.11$;
 - Hou–Xue–Zhang *q*-factor: -0.08 , $t = -0.27$;
 - Hou et al. *q*5: -0.01 , $t = -0.02$;
 - Stambaugh–Yuan mispricing: 0.07 , $t = 0.25$;
 - Daniel–Hirshleifer–Sun: 0.31 , $t = 0.87$.

Table 7
Portfolios sorted on conditional bias: So (2013) versus Binsbergen, Han, and Lopez-Lira (2023)

Strategy	Specification	Full	Subsample		Pre-Post	<i>p</i> -value
		Sample	Pre	Post		
BHL	Original	−1.46 (−5.11)	−1.67 (−4.52)	−1.30 (−3.09)	−0.37	0.51
	Replication with	−1.59	−1.82	−1.42	−0.40	0.53
	look-ahead bias	(−5.08)	(−4.63)	(−3.14)		
	Replication w/o	−0.25	−0.52	−0.06	−0.46	0.47
So (2013)	look-ahead bias	(−0.80)	(−1.35)	(−0.13)		
	BHL's replication	−0.67 (−4.05)	−1.11 (−3.79)	−0.33 (−1.79)	−0.78	0.03

This table reports average returns for value-weighted monthly rebalanced long-short portfolios formed on various measures of the earnings forecast bias. The estimates on rows 1 and 4 are from (Binsbergen, Han, and Lopez-Lira 2023, Table A13); those on rows 2 and 3 are from our replication of BHL with and without the look-ahead bias. We report the estimates for the full, pre-2000, and post-2000 samples. The *t*-values use White's (1980) heteroskedasticity robust standard errors as in BHL. The two rightmost columns test the difference in average returns between the pre- and post-2000 subperiods.

- The result is not a matter of choosing FF5. The corrected strategy does not earn robust abnormal returns across modern factor benchmarks.

5.9 Table 7: comparison with So (2013)

Read:

- BHL argue that their machine-learning strategy improves on So's earlier analyst-error strategy.
- With the look-ahead bias, the BHL-style strategy earns about −1.59% per month in the full sample.
- Without the look-ahead bias, it earns only −0.25%, with $t = -0.80$.
- The replicated So (2013) strategy earns about −0.67% with $t = -4.05$ in the full sample.
- Therefore, BHL improves on So only because of the look-ahead bias. Once corrected, the BHL strategy is weaker than So's strategy.

5.10 Look-ahead diagnostic regressions

Read:

- The paper regresses horizon- h actual earnings on BHL's horizon- h and horizon- $h + 1$ machine forecasts, controlling for analyst forecasts.
- If the horizon- $h + 1$ machine forecast contains horizon- h realized earnings, its coefficient should be positive.
- Using BHL's provided forecasts, $\hat{\beta}_2$ is strongly positive, with t -statistics as high as 30.2.
- The same pattern appears in the authors' look-ahead replication and disappears in the corrected replication.
- This is the most direct evidence that the original forecast output contains future realized earnings.

5.11 Other BHL results: equity issuance and anomaly returns

Read:

- Equity issuance results are not very sensitive to the look-ahead correction, but the interpretation changes.
- Conditional bias equals analyst forecast minus expected earnings. The paper shows that the expected-earnings component, not the analyst-bias component, drives the equity issuance relation.
- Similarly, the result that anomaly short legs are associated with high conditional bias appears to be driven by forecasted earnings rather than by analyst optimism.
- The result that anomalies are stronger among high-bias stocks remains meaningful and is not very sensitive to the look-ahead correction. In that application, the random-forest bias measure does somewhat better than the linear model: roughly $t = 4.66$ versus $t = 3.96$.

5.12 Alternative machine-learning forecasts

Read:

- The paper studies OLS, partial least squares, LASSO, elastic net, random forest, LightGBM, and a composite forecast.
- The composite forecast is most accurate overall, consistent with the general value of ensembles.
- However, analyst biases measured relative to these alternative benchmarks do not significantly predict returns.
- Better earnings forecasts and tradable alphas are different objects. Improving forecast accuracy is not sufficient to generate abnormal returns.

6 Short summary

- **Method.** Replicate BHL’s random-forest earnings forecasts and bias-sorted trading strategy, then correct the timing of the realized-earnings feature.
- **Identification.** The key comparison holds the pipeline fixed and changes only whether the “last-period realized earnings” feature is future information. A separate diagnostic regression directly tests whether longer-horizon forecasts contain shorter-horizon actual earnings.
- **Application.** U.S. stocks with analyst coverage, five forecast horizons, 1984–2019 forecasting sample and 1986–2019 return tests.
- **Main result.** The BHL alpha disappears once the look-ahead bias is removed. The five-factor alpha falls from about -1.71% per month with the bias to -0.41% without it.
- **Interpretation.** Machine learning is not the problem; timing is. Nonlinear methods cannot be evaluated as implementable forecasts unless every feature is observable at the forecast date.

7 Conclusion

7.1 Core conclusion

- Zhang, Zhu, and Linnainmaa show that BHL’s strongest findings are driven by a look-ahead bias in the realized-earnings feature.
- Once the feature timing is corrected, the machine-learning benchmark no longer generates a statistically significant alpha under modern factor models.
- Linear models forecast earnings about as accurately as the corrected random forest, and alternative machine-learning models do not restore the original alpha.

7.2 Economic intuition

- A future realized earnings variable is highly informative about future EPS by construction.
- If that future variable enters a machine-learning forecast, the benchmark appears unusually accurate.
- When analyst forecasts are compared with this contaminated benchmark, the estimated “bias” partly reflects information that analysts and investors could not have known.
- Sorting stocks on this contaminated bias can therefore look like a strong anomaly even though the signal is not implementable.

7.3 Takeaway

- The paper is a cautionary replication study for empirical finance and machine learning.
- The main lesson is that machine-learning models do not relax the usual econometric discipline about information sets. They make that discipline more important because feature engineering across horizons can quietly introduce future information.
- The corrected evidence leaves open the possibility that analyst expectations and their term structure matter for returns, but it rejects the claim that BHL’s random-forest conditional bias generates a large robust tradable alpha.
- The authors also provide replication code through Harvard Dataverse: <https://doi.org/10.7910/DVN/YWZJOP>.